

# Nitin Kumar Shukla

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## Summary

Integrated M.Tech student in AI at VIT specializing in scalable machine learning systems and full-stack software development. Highly experienced in automating complex workflows through advanced OCR and seamless API integrations. Proficient in Python, Java, and AWS, with a solid foundation in the complete SDLC. Proven track record of translating complex data into actionable insights to build high-quality technical solutions.

## Education

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| <b>Vellore Institute of Technology (VIT)</b><br><i>Integrated M.Tech in Artificial Intelligence</i> | 2022 – 2027<br>CGPA: 8.14 |
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## Experience

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| <b>South Eastern Coalfields Limited (SECL)</b><br><i>Data Analyst Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Oct 2024 – Dec 2024<br><i>Bilaspur, India</i> |
| <ul style="list-style-type: none"><li>Automated reimbursement workflows for 500+ daily claims by developing a complete medical bill scrutiny system.</li><li>Engineered advanced OCR pipelines using Tesseract and OpenCV to extract structured data from medical records.</li><li>Reduced processing errors by 30% and flagged high-risk claims by designing Python anomaly detection modules.</li><li>Optimized backend system performance by utilizing RESTful APIs for real-time data retrieval and processing.</li></ul> |                                               |
| <b>AI Developer at FloorWalk(Part-Time)</b><br><i>Automated Audit Verification System</i>                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mar 2024 – Present<br><i>Remote</i>           |
| <ul style="list-style-type: none"><li>Architected autonomous AI agents using LangGraph and LangChain to seamlessly automate complex audit verifications.</li><li>Leveraged Generative AI and RAG pipelines to extract, validate, and cross-reference compliance data from large reports.</li></ul>                                                                                                                                                                                                                            |                                               |

## Technical Skills

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| <b>Programming:</b> Java, Python (Advanced), C++, SQL (PostgreSQL/MySQL), JavaScript, RESTful APIs                  |
| <b>Machine Learning:</b> Generative AI, LangChain, LangGraph, TensorFlow, PyTorch, Scikit-learn, XGBoost, Pandas    |
| <b>Cloud &amp; Tools:</b> AWS (EC2, S3, Lambda), Kubernetes, Docker, Git/GitHub, Linux/Unix Shell Scripting         |
| <b>Core Concepts:</b> Data Structures & Algorithms, SDLC, Statistics, System Design, Cross-Functional Collaboration |

## Projects

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| <b>Stock Market Prediction System</b>   <i>Python, Scikit-learn, XGBoost, Matplotlib</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Jan 2024  |
| <ul style="list-style-type: none"><li>Designed an end-to-end ML pipeline ingesting OHLCV time-series data, computing 20+ technical indicators (RSI, MACD, Bollinger Bands) via Pandas/TA-Lib, and engineering lag features with rolling-window transformations for temporal pattern extraction.</li><li>Implemented hyperparameter tuning via GridSearchCV and cross-validated ensemble stacking of Random Forest and XGBoost classifiers, reducing overfitting through L1/L2 regularization and achieving an F1-score of 0.83 on unseen market regimes.</li></ul> |           |
| <b>Deepfake Detection Platform</b>   <i>TensorFlow, CNN, OpenCV, Flask</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sept 2023 |
| <ul style="list-style-type: none"><li>Architected a ResNet-50 based CNN with custom attention layers to classify deepfake frames by detecting spatial inconsistencies, GAN-based blending artifacts, and frequency-domain anomalies using FFT preprocessing.</li><li>Deployed a scalable Flask REST API with OpenCV-powered video frame sampling pipeline, integrating MTCNN for multi-face detection and alignment, reducing inference latency by optimizing batch prediction throughput.</li></ul>                                                               |           |

## Certifications

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|---------------------------------------------------------------------------|------|
| <b>Kubernetes for Absolute Beginners</b>   <i>Packt</i>                   | 2025 |
| <b>Applied Machine Learning in Python</b>   <i>University of Michigan</i> | 2023 |
| <b>Cloud Computing</b>   <i>NPTEL</i>                                     | 2023 |